E1 244 Detection and Estimation Theory
Assignment 1

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Problem 1 (6.3)

Ans: The shifted exponential distribution is given by:

$$f(x|\mu,\sigma) = \frac{1}{\sigma} e^{-\frac{(x-\mu)}{\sigma}}, \quad \mu < x < \infty, \quad 0 < \sigma < \infty$$

This can be written as:

$$f(x|\mu,\sigma) = \frac{1}{\sigma} e^{-\frac{(x-\mu)}{\sigma}} \cdot \mathbf{1}\{\mu < x < \infty\}$$

Let $\mathbf{X} = (X_1, X_2, \ldots, X_n)$ and $\boldsymbol{\theta} = \begin{bmatrix} \mu \\ \sigma \end{bmatrix}$.

The joint pdf is:

$$\begin{aligned} P_\theta(\mathbf{X}) &= \prod_{i=1}^{n} f(x = X_i|\mu,\sigma) \\ &= \frac{1}{\sigma^n} \prod_{i=1}^{n} e^{-\frac{(X_i-\mu)}{\sigma}} \cdot \mathbf{1}\{\mu < X_i < \infty\} \\ &= \frac{1}{\sigma^n} e^{-\frac{(\sum X_i - n\mu)}{\sigma}} \cdot \mathbf{1}\{\mu < \min_i(X_i) < \infty\} \end{aligned}$$

By the **Neyman-Fisher Factorization Theorem**:
If $P_\theta(\mathbf{X}) = g(T(\mathbf{X}), \boldsymbol{\theta}) \cdot h(\mathbf{X})$, then $T(\mathbf{X})$ is a sufficient statistic for $\boldsymbol{\theta}$.
Here, we can identify:

$$g(T(\mathbf{X}), \boldsymbol{\theta}) = \frac{1}{\sigma^n} e^{-\frac{(\sum X_i - n\mu)}{\sigma}} \cdot \mathbf{1}\{\mu < \min_i(X_i) < \infty\}$$

and

$$h(\mathbf{X}) = 1$$

Therefore, the **sufficient statistic** is:

$$\boxed{T(\mathbf{X}) = \begin{bmatrix} \min_i(X_i) \\ \sum_i X_i \end{bmatrix}}$$

Problem 2 (6.9)

Ans: (a) The pdf is:

$$f(x|\theta) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\theta)^2}{2}}$$

Let $\mathbf{X} = (X_1, X_2, \ldots, X_n)$ and $\mathbf{Y} = (Y_1, Y_2, \ldots, Y_n)$ be two data points.

$$\begin{aligned} P_\theta(\mathbf{X}) &= \prod_{i=1}^{n} f(x = X_i|\theta) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-\frac{(X_i-\theta)^2}{2}} \\ &= \frac{1}{(2\pi)^{n/2}} e^{-\frac{n}{2}(\bar{X}-\theta)^2} \cdot e^{-\frac{1}{2}\sum(X_i-\bar{X})^2} \end{aligned}$$

Similarly:

$$P_\theta(\mathbf{Y}) = \frac{1}{(2\pi)^{n/2}} e^{-\frac{n}{2}(\bar{Y}-\theta)^2} \cdot e^{-\frac{1}{2}\sum(Y_i-\bar{Y})^2}$$

The likelihood ratio is:

$$\begin{aligned} \frac{P_\theta(\mathbf{X})}{P_\theta(\mathbf{Y})} &= e^{-\frac{n}{2}[(\bar{X}-\theta)^2-(\bar{Y}-\theta)^2]} \cdot e^{-\frac{1}{2}[\sum(X_i-\bar{X})^2-\sum(Y_i-\bar{Y})^2]} \\ &= e^{-\frac{n}{2}[\bar{X}^2-\bar{Y}^2-2\theta(\bar{X}-\bar{Y})]} \cdot e^{-\frac{1}{2}[\sum(X_i-\bar{X})^2-\sum(Y_i-\bar{Y})^2]} \end{aligned}$$

If this is a constant function of θ, then:

$$\Rightarrow 2\theta(\bar{X} - \bar{Y}) = 0$$

$$\Rightarrow \bar{X} - \bar{Y} = 0 \Rightarrow \bar{X} = \bar{Y}$$

And if $\bar{X} = \bar{Y}$, then:

$$\frac{P_\theta(\mathbf{X})}{P_\theta(\mathbf{Y})} = e^{-\frac{1}{2}[\sum(X_i-\bar{X})^2-\sum(Y_i-\bar{Y})^2]}$$

which is a constant function of θ.

So, $\boxed{\bar{X}}$ **is a minimal sufficient statistic for** θ.

(d) Let $\mathbf{X} = (X_1, X_2, \ldots, X_n)$ and $\mathbf{Y} = (Y_1, Y_2, \ldots, Y_n)$ be two random samples.

Then:

$$P_\theta(\mathbf{X}) = \prod_{i=1}^{n} f(x = X_i|\theta) = \prod_{i=1}^{n} \frac{1}{\pi[1 + (X_i - \theta)^2]} = \frac{1}{\pi^n} \cdot \frac{1}{\prod_{i=1}^{n}(1 + (X_i - \theta)^2)}$$

Similarly:

$$P_\theta(\mathbf{Y}) = \frac{1}{\pi^n} \cdot \frac{1}{\prod_{i=1}^{n}(1 + (Y_i - \theta)^2)}$$

So:

$$\frac{P_\theta(\mathbf{X})}{P_\theta(\mathbf{Y})} = \frac{\prod_{i=1}^{n}(1 + (Y_i - \theta)^2)}{\prod_{i=1}^{n}(1 + (X_i - \theta)^2)}$$

The numerator and denominator are functions of θ. If $\frac{P_\theta(\mathbf{X})}{P_\theta(\mathbf{Y})}$ is independent of θ, then all the coefficients in the numerator and denominator must be the same.

$$\Rightarrow X_i = Y_i \quad \forall i = 1, 2, \ldots, n$$

And if $X_i = Y_i$ $\forall i = 1, 2, \ldots, n$, then $\frac{P_\theta(\mathbf{X})}{P_\theta(\mathbf{Y})}$ is independent of θ.

So, $\boxed{T(\mathbf{X}) = (X_1, X_2, \ldots, X_n)}$ **is the minimal sufficient statistic for** θ.

(e) Let $\mathbf{X} = (X_1, X_2, \ldots, X_n)$ and $\mathbf{Y} = (Y_1, Y_2, \ldots, Y_n)$ be two random samples.

$$P_\theta(\mathbf{X}) = \prod_{i=1}^{n} f(x = X_i|\theta) = \frac{1}{2^n} e^{-\sum_i |X_i - \theta|}$$

Similarly:

$$P_\theta(\mathbf{Y}) = \frac{1}{2^n} e^{-\sum_i |Y_i - \theta|}$$

$$\frac{P_\theta(\mathbf{X})}{P_\theta(\mathbf{Y})} = e^{-\sum_i \{|X_i - \theta| - |Y_i - \theta|\}}$$

This term will be independent of θ only when:

$$\text{number of } X_i < \theta = \text{number of } Y_i < \theta \quad \forall \theta \in \Theta$$

It is only possible when $X_{(i)} = Y_{(i)}$, i.e., the order statistics are the same.

The reverse side: $X_{(i)} = Y_{(i)} \Rightarrow e^{-\sum_i \{|X_i-\theta|-|Y_i-\theta|\}}$ is independent of θ (trivial).

So, $\boxed{T(\mathbf{X}) = \{X_{(1)}, X_{(2)}, \ldots, X_{(n)}\}}$ **is the minimal sufficient statistic**.

Problem 3 (6.12)

Ans: (a) Given $P_\theta(X|N=n) = \binom{n}{X}\theta^X(1-\theta)^{n-X}$

For another observation (Y, M), we get the distribution as:

$$P_\theta(Y|M=m) = \binom{m}{Y}\theta^Y(1-\theta)^{m-Y}$$

$$\frac{P_\theta(X|N=n)}{P_\theta(Y|M=m)} = \frac{\binom{n}{X}}{\binom{m}{Y}} \cdot \frac{\theta^X(1-\theta)^{n-X}}{\theta^Y(1-\theta)^{m-Y}}$$

This is independent of θ only when $m = n$ and $X = Y$.

Reverse is trivially true.

So, $\boxed{(X, N)}$ **is the minimal sufficient statistic**.

Distribution of N is fixed. So N is **ancillary** for θ.

(b) $P_\theta(X, N) = P_\theta(X|N) \cdot P_\theta(N) = \binom{N}{X}\theta^X(1-\theta)^{N-X} \cdot p_N$

Now:

$$\begin{aligned}
E_\theta\left(\frac{X}{N}\right) &= \sum_{N\geq 1}\sum_{X=0}^{N} \frac{X}{N} \cdot \binom{N}{X}\theta^X(1-\theta)^{N-X} \cdot p_N \\
&= \sum_{N\geq 1} \frac{p_N}{N} \cdot \sum_{X=1}^{N} X \cdot \frac{N!}{X!(N-X)!} \cdot \theta^X(1-\theta)^{N-X} \\
&= \sum_{N\geq 1} \frac{p_N}{N} \cdot \sum_{X=1}^{N} N \cdot \binom{N-1}{X-1} \cdot \theta^X(1-\theta)^{N-X} \\
&= \sum_{N\geq 1} \frac{p_N}{N} \cdot N \cdot \theta \cdot \sum_{X=1}^{N} \binom{N-1}{X-1} \cdot \theta^{X-1}(1-\theta)^{N-X}
\end{aligned}$$

Define $Y = X - 1$, then:

$$E_\theta\left(\frac{X}{N}\right) = \sum_{N\geq 1} p_N \cdot \theta \cdot \sum_{Y=0}^{N-1} \binom{N-1}{Y} \cdot \theta^Y (1-\theta)^{N-1-Y}$$
$$= \sum_{N\geq 1} p_N \cdot \theta = \theta \cdot \sum_{N\geq 1} p_N = \theta$$

So, $\boxed{\frac{X}{N}}$ **is the unbiased estimator of** θ.

(c) $\text{Var}\left(\frac{X}{N}\right) = E\left(\frac{X}{N}\right)^2 - \left[E\left(\frac{X}{N}\right)\right]^2$
Now, $E\left(\frac{X^2}{N^2}\right) = E\left(\frac{X(X-1)}{N^2} + \frac{X}{N^2}\right)$

$$E\left(\frac{X(X-1)}{N^2}\right) = \sum_{N\geq 1} \frac{p_N}{N^2} \cdot \sum_{X=2}^{N} X(X-1) \cdot \frac{N!}{X!(N-X)!} \cdot \theta^X (1-\theta)^{N-X}$$
$$= \sum_{N\geq 1} \frac{p_N}{N^2} \cdot N(N-1) \cdot \theta^2 \cdot \sum_{X=2}^{N} \binom{N-2}{X-2} \cdot \theta^{X-2}(1-\theta)^{N-X}$$
$$= \theta^2 \sum_{N\geq 1} \frac{p_N}{N^2} \cdot N(N-1) \cdot (\theta + 1 - \theta)^{N-2} \quad \text{[By similar substitution]}$$
$$= \theta^2 \sum_{N\geq 1} \frac{p_N(N-1)}{N}$$

Similarly, $E\left(\frac{X}{N^2}\right) = \theta \sum_{N\geq 1} \frac{p_N}{N}$
So, $E\left(\frac{X^2}{N^2}\right) = \sum_{N\geq 1} \frac{p_N}{N} \left[\theta^2(N-1) + \theta\right]$
Now:

$$\text{Var}\left(\frac{X}{N}\right) = \sum_{N\geq 1} \frac{p_N}{N} \left[\theta^2(N-1) + \theta\right] - \theta^2$$
$$= \sum_{N\geq 1} \frac{p_N}{N} \left[\theta^2(N-1) + \theta\right] - \theta^2 \sum_{N\geq 1} \frac{p_N \cdot N}{N}$$
$$= \sum_{N\geq 1} \frac{p_N}{N} \left[\theta - \theta^2\right] = \theta(1-\theta) \sum_{N\geq 1} \frac{p_N}{N}$$

$$\boxed{\text{Var}\left(\frac{X}{N}\right) = \theta(1-\theta) \cdot E\left(\frac{1}{N}\right)}$$

Problem 4 (6.18)

Ans: Let $Y = X_1 + X_2 + \cdots + X_n$ where $X_i \sim \text{Poisson}(\lambda)$ i.i.d.

$$P(X_i = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \ldots$$

The moment generating function of X_i is:

$$\begin{aligned} M_X(t) = E(e^{tX}) &= \sum_{x \geq 0} e^{tx} \cdot \frac{\lambda^x e^{-\lambda}}{x!} \\ &= \sum_{x \geq 0} \frac{(\lambda e^t)^x \cdot e^{-\lambda}}{x!} \\ &= e^{-\lambda} \sum_{x \geq 0} \frac{(\lambda e^t)^x}{x!} \\ &= e^{-\lambda} \cdot e^{\lambda e^t} = e^{\lambda(e^t - 1)} \end{aligned}$$

$$\begin{aligned} M_Y(t) = E(e^{tY}) &= E(e^{t(X_1 + X_2 + \cdots + X_n)}) \\ &= E(e^{tX_1}) \cdot E(e^{tX_2}) \cdots E(e^{tX_n}) \\ &= e^{\lambda(e^t - 1)} \cdot e^{\lambda(e^t - 1)} \cdots e^{\lambda(e^t - 1)} \\ &= e^{n\lambda(e^t - 1)} \end{aligned}$$

So, $Y \sim \text{Poisson}(n\lambda)$.
Therefore, $T = \sum_i X_i \sim \text{Poisson}(n\lambda)$.
Now, if $E_\lambda(g(T)) = 0 \ \forall \lambda$:

$$\begin{aligned} E_\lambda(g(T)) &= \sum_{t=0}^{\infty} g(t) \cdot \frac{(n\lambda)^t \cdot e^{-n\lambda}}{t!} = 0 \quad \forall \lambda \\ &= e^{-n\lambda} \sum_{t=0}^{\infty} \frac{g(t) \cdot (n\lambda)^t}{t!} = 0 \quad \forall \lambda \end{aligned}$$

Let $\frac{g(t) \cdot n^t}{t!} = \alpha(t)$.
So, $\sum_{t=0}^{\infty} \alpha(t) \cdot \lambda^t = 0 \quad \forall \lambda$
Since it is a polynomial in λ, all coefficients must be 0.

$$\Rightarrow \alpha(t) = 0 \Rightarrow g(t) = 0 \quad \forall t$$

So, the family of distribution of $\sum X_i$ is complete.

Problem 5 (6.19)

Ans: Distribution 1:

Let $T \sim$ Dist 1, i.e., $P(T=0)=p$, $P(T=1)=3p$, $P(T=2)=1-4p$ for $0<p<1/4$.

If $E_p(g(T)) = g(0)\cdot p + g(1)\cdot 3p + g(2)\cdot(1-4p) = 0$

$$= \{g(0)+3g(1)-4g(2)\}\cdot p + g(2) = 0$$

We can find $g(0)=1$, $g(1)=-\frac{1}{3}$, $g(2)=0$ for all $0<p<1/4$.

In that case $\forall p$: $E_p(g(T))=0$ but $g(T)\neq 0\ \forall t$.

So, Distribution 1 is NOT complete.

Distribution 2:

Let $T \sim$ Dist 2, i.e., $P(T=0)=p$, $P(T=1)=p^2$, $P(T=2)=1-p-p^2$ for $0<p<1/2$.

If $E_p(g(T)) = g(0)\cdot p + g(1)\cdot p^2 + (1-p-p^2)\cdot g(2) = 0$

$$g(2) + \{g(0)-g(2)\}\cdot p + \{g(1)-g(2)\}\cdot p^2 = 0$$

All the coefficients must be 0 for all $0<p<1/2$:

$$\begin{aligned} g(2) &= 0 \\ g(0)-g(2) &= 0 \\ g(1)-g(2) &= 0 \end{aligned}$$

$$\Rightarrow g(0)=0, \quad g(1)=0, \quad g(2)=0$$

So, Distribution 2 family of X is complete.

Problem 6 (6.30)

Ans: (a) Let $X_{(1)} = \min_i X_i$ and $\mathbf{X} = (X_1, X_2, \ldots, X_n)$.

$$
\begin{aligned}
P_\theta(\mathbf{X}) &= \prod_{i=1}^{n} f(x = X_i|\mu) \\
&= \prod_{i=1}^{n} e^{-(X_i-\mu)} \cdot \mathbf{1}\{\mu < X_i\} \\
&= e^{-(\sum X_i - n\mu)} \cdot \mathbf{1}\{\mu < \min_i(X_i)\} \\
&= \underbrace{e^{-\sum X_i}}_{h(\mathbf{X})} \cdot \underbrace{e^{n\mu} \cdot \mathbf{1}\{\mu < X_{(1)}\}}_{g(T(\mathbf{X}),\theta)}
\end{aligned}
$$

So, by **Neyman-Fisher Factorization**:
$T(\mathbf{X}) = X_{(1)}$ is a **sufficient statistic**.

Finding the distribution of $X_{(1)}$:
$X_{(1)} = \min\{X_1, X_2, \ldots, X_n\}$

$$
\begin{aligned}
P(X_{(1)} < t) &= 1 - P(X_{(1)} > t) \\
&= 1 - P(\min_i\{X_i\} > t) \\
&= 1 - P(X_1 > t) \cdot P(X_2 > t) \cdots P(X_n > t)
\end{aligned}
$$

Now:

$$
\begin{aligned}
P(X_i < t) &= \int_{-\infty}^{t} e^{-(x-\mu)} \cdot \mathbf{1}\{\mu < x\} dx = \int_{\mu}^{t} e^{-(x-\mu)} dx \\
&= \int_{0}^{t-\mu} e^{-y} dy \quad [\text{taking } y = x - \mu] \\
&= \left[-e^{-y}\right]_0^{t-\mu} = 1 - e^{-(t-\mu)}, \quad t > \mu
\end{aligned}
$$

Therefore:

$$P(X_{(1)} < t) = 1 - e^{-(t-\mu)n}, \quad t > \mu$$

The pdf is:

$$p_\mu(X_{(1)}) = \frac{d}{dt}\left\{1 - e^{-(t-\mu)n}\right\} = n \cdot e^{\mu n} \cdot e^{-nt} = n \cdot e^{-(t-\mu)n}, \quad t > \mu$$

Proving completeness:

Now, if $E_\mu(g(T)) = 0 \ \forall \mu \in \mathbb{R}$:

$$\begin{aligned} E_\mu(g(T)) &= \int_{-\infty}^{\infty} g(t) \cdot n \cdot e^{-(t-\mu)n} \cdot \mathbf{1}\{t > \mu\} dt \\ &= \int_{\mu}^{\infty} g(t) \cdot n \cdot e^{-(t-\mu)n} dt = 0 \quad \forall \mu \in \mathbb{R} \end{aligned}$$

Take $u = t - \mu$:

$$\Rightarrow \int_0^{\infty} g(u+\mu) \cdot n \cdot e^{-nu} du = 0$$

$\Rightarrow$ Convolution with exponential kernel is 0.
$\Rightarrow$ g is zero $\forall t$.
$\Rightarrow$ $X_{(1)}$ **is a complete statistic.**

(b) Let $Y \sim \exp(1)$, i.e., $f(y) = e^{-y}$, $0 < y < \infty$.
Now, $X = Y + \mu$ (shifted by constant μ).

$$S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2 = \frac{1}{n-1} \sum (Y_i - \bar{Y})^2$$

Note that $\sum (Y_i - \bar{Y})^2$ does not depend on μ.
So, S^2 is an **ancillary statistic**.
So, by **Basu's Theorem**: $\boxed{X_{(1)} \text{ and } S^2 \text{ are independent}}$.

Problem 7 (6.34)

Ans: Experiment 1: N chosen randomly, then N Bernoulli trials were done.
Experiment 2: N is fixed at n.
Sample space: $\mathcal{Y}_1 = \{0, 1, 2, \ldots\}$, $\mathcal{Y}_2 = \{0, 1, 2, \ldots, n\}$

For Experiment 1:

$$P_\theta^{(1)}(X, N) = p_N \cdot \binom{N}{X} \theta^X (1-\theta)^{N-X}$$

For Experiment 2:

$$P_\theta^{(2)}(X) = \binom{n}{X} \theta^X (1-\theta)^{n-X}$$

For $N = n$:

$$P_\theta^{(1)}(X, N = n) = p_n \cdot \binom{n}{X} \theta^X (1-\theta)^{n-X} \propto P_\theta^{(2)}(X)$$

So, the likelihood is the same (up to a constant that doesn't depend on θ).

Therefore, if N is chosen to be n, it does not matter whether it was chosen randomly or not. The inference about θ based on the likelihood will be identical in both cases.

Problem 6 (Continued)

Ans: (b) Let $y \sim \mathcal{N}(s, R(\theta))$ where s is a known vector and $R(\theta)$ is the covariance matrix that depends on θ.

The pdf is:

$$f_\theta(y) = \frac{1}{(2\pi)^{n/2}|R(\theta)|^{1/2}} \exp\left(-\frac{1}{2}(y-s)^T R^{-1}(\theta)(y-s)\right)$$

The log-likelihood is:

$$\ln f_\theta(y) = -\frac{n}{2}\ln(2\pi) - \frac{1}{2}\ln|R(\theta)| - \frac{1}{2}(y-s)^T R^{-1}(\theta)(y-s)$$

Using the matrix calculus identities:

- $\frac{\partial}{\partial\theta_j}\ln|R| = \text{Trace}\left(R^{-1}\frac{\partial R}{\partial\theta_j}\right)$
- $\frac{\partial R^{-1}}{\partial\theta_j} = -R^{-1}\frac{\partial R}{\partial\theta_j}R^{-1}$

Taking the derivative with respect to θ_j:

$$\frac{\partial \ln f_\theta(y)}{\partial\theta_j} = -\frac{1}{2}\text{Trace}\left(R^{-1}\frac{\partial R}{\partial\theta_j}\right) + \frac{1}{2}(y-s)^T R^{-1}\frac{\partial R}{\partial\theta_j}R^{-1}(y-s)$$

Let $z = y - s \sim \mathcal{N}(0, R(\theta))$ and define $A_j = R^{-1}\frac{\partial R}{\partial\theta_j}$. Then:

$$\frac{\partial \ln f_\theta(y)}{\partial\theta_j} = \frac{1}{2}\left[z^T A_j R^{-1} z - \text{Trace}(A_j)\right] = \frac{1}{2}\text{Trace}\left(A_j(R^{-1}zz^T - I)\right)$$

Using the Fisher information formula $[I_y(\theta)]_{jk} = E\left[\frac{\partial \ln f}{\partial \theta_j}\frac{\partial \ln f}{\partial \theta_k}\right]$ and the property for Gaussian random vectors that:

$$E[\text{Trace}(A_j(R^{-1}zz^T - I))\text{Trace}(A_k(R^{-1}zz^T - I))] = 2\text{Trace}(A_jA_k)$$

We get:

$$\boxed{[I_y(\theta)]_{jk} = \frac{1}{2}\text{Trace}\left(R^{-1}(\theta)\frac{\partial R(\theta)}{\partial \theta_j}R^{-1}(\theta)\frac{\partial R(\theta)}{\partial \theta_k}\right)}$$

(c) Let $y \sim \mathcal{N}(s(\theta), R(\theta))$ where both mean and covariance depend on θ. The log-likelihood is:

$$\ln f_\theta(y) = -\frac{n}{2}\ln(2\pi) - \frac{1}{2}\ln|R(\theta)| - \frac{1}{2}(y - s(\theta))^T R^{-1}(\theta)(y - s(\theta))$$

Taking the derivative:

$$\frac{\partial \ln f}{\partial \theta_j} = \underbrace{\left(\frac{\partial s}{\partial \theta_j}\right)^T R^{-1}(y - s)}_{\text{from mean}} + \underbrace{\left[-\frac{1}{2}\text{Trace}\left(R^{-1}\frac{\partial R}{\partial \theta_j}\right) + \frac{1}{2}(y - s)^T R^{-1}\frac{\partial R}{\partial \theta_j}R^{-1}(y - s)\right]}_{\text{from covariance}}$$

When computing the Fisher information, the cross terms between the mean-dependent and covariance-dependent parts vanish because:

- The mean-dependent term is linear in $(y - s)$, which has zero mean
- The covariance-dependent terms are either constant or quadratic in $(y-s)$
- $E[(y - s)] = 0$ implies the cross-product expectation vanishes

Therefore, the Fisher information matrix is **additive**:

$$\boxed{[I_y(\theta)]_{jk} = \left[\frac{\partial s(\theta)}{\partial \theta_j}\right]^T R^{-1}\left[\frac{\partial s(\theta)}{\partial \theta_k}\right] + \frac{1}{2}\text{Trace}\left(R^{-1}\frac{\partial R}{\partial \theta_j}R^{-1}\frac{\partial R}{\partial \theta_k}\right)}$$

This is the sum of the Fisher information matrices from parts (a) and (b).

Problem 7 (Continued)

Ans: (b) From part (a), we have $y_k \sim \mathcal{N}(0, \theta s_k^2 + 1)$ independently.
The log-likelihood is:

$$\ln p_\theta(y) = \sum_{k=1}^{N} \left[-\frac{1}{2}\ln(2\pi) - \frac{1}{2}\ln(\theta s_k^2 + 1) - \frac{y_k^2}{2(\theta s_k^2 + 1)} \right]$$

Taking derivatives:

$$\frac{\partial \ln p_\theta(y)}{\partial \theta} = \sum_{k=1}^{N} \left[-\frac{s_k^2}{2(\theta s_k^2 + 1)} + \frac{s_k^2 y_k^2}{2(\theta s_k^2 + 1)^2} \right]$$

$$\frac{\partial^2 \ln p_\theta(y)}{\partial \theta^2} = \sum_{k=1}^{N} \left[\frac{s_k^4}{2(\theta s_k^2 + 1)^2} - \frac{s_k^4 y_k^2}{(\theta s_k^2 + 1)^3} \right]$$

The Fisher Information is:

$$I_y(\theta) = -E\left[\frac{\partial^2 \ln p_\theta(y)}{\partial \theta^2} \right]$$

Since $E[y_k^2] = \theta s_k^2 + 1$:

$$\begin{aligned} I_y(\theta) &= -\sum_{k=1}^{N} \left[\frac{s_k^4}{2(\theta s_k^2 + 1)^2} - \frac{s_k^4(\theta s_k^2 + 1)}{(\theta s_k^2 + 1)^3} \right] \\ &= -\sum_{k=1}^{N} \left[\frac{s_k^4}{2(\theta s_k^2 + 1)^2} - \frac{s_k^4}{(\theta s_k^2 + 1)^2} \right] \\ &= \sum_{k=1}^{N} \frac{s_k^4}{2(\theta s_k^2 + 1)^2} \end{aligned}$$

Therefore, the **Cramer-Rao Bound** is:

$$\boxed{\mathrm{CRB}(\theta) = \frac{1}{I_y(\theta)} = \frac{1}{\sum_{k=1}^{N} \frac{s_k^4}{2(\theta s_k^2 + 1)^2}}}$$

(c) When $s_i \in \{+1, -1\}$, we have $s_i^2 = 1$ for all i.
So $y_k \sim \mathcal{N}(0, \theta + 1)$ independently for all k.
The log-likelihood becomes:

$$\ln p_\theta(y) = -\frac{N}{2}\ln(2\pi) - \frac{N}{2}\ln(\theta + 1) - \frac{1}{2(\theta + 1)} \sum_{k=1}^{N} y_k^2$$

Setting the derivative to zero:

$$\frac{\partial \ln p_\theta(y)}{\partial \theta} = -\frac{N}{2(\theta+1)} + \frac{\sum_{k=1}^{N} y_k^2}{2(\theta+1)^2} = 0$$

Solving:

$$\frac{N}{\theta+1} = \frac{\sum_{k=1}^{N} y_k^2}{(\theta+1)^2}$$

$$\Rightarrow \theta + 1 = \frac{1}{N}\sum_{k=1}^{N} y_k^2$$

Since we require $\theta \geq 0$, the **ML estimate** is:

$$\boxed{\hat{\theta}_{ML} = \max\left(0, \frac{1}{N}\sum_{k=1}^{N} y_k^2 - 1\right)}$$

For typical scenarios where the unconstrained estimate is positive:

$$\hat{\theta}_{ML} = \frac{1}{N}\sum_{k=1}^{N} y_k^2 - 1$$

(d) Let $\hat{\theta} = \frac{1}{N}\sum_{k=1}^{N} y_k^2 - 1$ (ignoring the boundary constraint for analysis).
Bias:

$$E[\hat{\theta}] = \frac{1}{N}\sum_{k=1}^{N} E[y_k^2] - 1 = \frac{1}{N} \cdot N \cdot (\theta+1) - 1 = \theta$$

$\Rightarrow$ **Bias** $= E[\hat{\theta}] - \theta = 0$ (unbiased estimator)
Variance:
For $y_k \sim \mathcal{N}(0, \sigma^2)$ where $\sigma^2 = \theta + 1$:

$$\mathrm{Var}(y_k^2) = E[y_k^4] - (E[y_k^2])^2 = 3\sigma^4 - \sigma^4 = 2\sigma^4 = 2(\theta+1)^2$$

Since the y_k are independent:

$$\mathrm{Var}(\hat{\theta}) = \mathrm{Var}\left(\frac{1}{N}\sum_{k=1}^{N} y_k^2\right) = \frac{1}{N^2}\sum_{k=1}^{N} \mathrm{Var}(y_k^2) = \frac{N \cdot 2(\theta+1)^2}{N^2} = \frac{2(\theta+1)^2}{N}$$

Comparison with CRB:

From part (b) with $s_k^2 = 1$:

$$I_y(\theta) = \sum_{k=1}^{N} \frac{1}{2(\theta+1)^2} = \frac{N}{2(\theta+1)^2}$$

$$\text{CRB} = \frac{1}{I_y(\theta)} = \frac{2(\theta+1)^2}{N}$$

Conclusion:

$$\boxed{\text{Var}(\hat{\theta}_{ML}) = \frac{2(\theta+1)^2}{N} = \text{CRB}}$$

The ML estimator achieves the Cramer-Rao Bound with equality, so it is an **efficient estimator**. Combined with being unbiased, the ML estimator $\hat{\theta}_{ML} = \frac{1}{N}\sum_{k=1}^{N} y_k^2 - 1$ is the **MVU estimator** for the case when $s_i \in \{+1, -1\}$.

Problem 8

Ans: Let $y_k \sim$ i.i.d. Unif$(0, \theta)$ for $k = 1, 2, \ldots, N$.

Consider the two estimates:

$$\hat{\theta}_1 = \max_{k=1}^{N}\{y_k\} \quad \text{and} \quad \hat{\theta}_2 = \frac{2}{N}\sum_{k=1}^{N} y_k$$

(a) For $N = 10,000$ and $\theta = 1, 2, 3, 4$:

Discussion: The Max estimator ($\hat{\theta}_1$) performs better than the 2×Mean estimator ($\hat{\theta}_2$) for all values of θ. This was obvious as discussed in the class, the max estimator is a sufficient statistics while the twice of mean statistics is not. And sufficient statistics captures all the information effectively giving better estimation than a not sufficient statistics.

(b) For $\theta = 1$ and $N = 10, 50, 100, 1000, 5000, 10000$:

Discussion: The estimation error decreases as N increases, showing a roughly linear trend on the semi-log plot. This is the golden rule of statistics. The idea mainly shown here is statistical regularity. As we increase the number of samples, the error of good estimator will converge to 0. Already, it is shown in the earlier problem that the max is a good estimator.